

Hospital Management System

SQL Database Project Documentation

This project is a complete relational database for managing hospital operations. It includes departments, doctors, patients, appointments, and billing records.

1. Project Overview

The database was built using MySQL. It shows how to design tables with Primary Keys and Foreign Keys, apply constraints, load realistic sample data, and write different types of SQL queries. The queries cover filtering, grouping, joins, and nested queries.

Database Name: hospital_management_system

Number of Tables: 5

Total Sample Records: About 485 rows

2. Data Summary

Table	Rows	Description
departments	50	Hospital departments such as Cardiology, Neurology, and Emergency
doctors	60	Doctors linked to departments, with specialization and salary
patients	120	Patient details including blood type, registration date, and contact info
appointments	150	Appointments between patients and doctors with status and reason
bills	105	Billing records linked to appointments and patients

3. Table Relationships

The tables are connected using Foreign Keys as shown below:

Child Table	Foreign Key	References
doctors	department_id	departments(department_id)
appointments	patient_id	patients(patient_id)
appointments	doctor_id	doctors(doctor_id)
bills	appointment_id	appointments(appointment_id)
bills	patient_id	patients(patient_id)

Flow: Departments link to Doctors. Doctors and Patients both link to Appointments. Appointments and Patients both link to Bills.

4. Table Structures

4.1 departments

Holds information about hospital departments.

Column	Type	Constraints	Description
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department_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each department
department_name	VARCHAR(100)	NOT NULL, UNIQUE	Name of the department
location	VARCHAR(100)	NOT NULL	Building and floor
phone_extension	VARCHAR(20)	Optional	Internal phone extension
annual_budget	DECIMAL(12,2)	Optional	Yearly budget amount
created_at	DATE	Optional	Date the record was created

4.2 doctors

Stores details about doctors including their department and salary.

Column	Type	Constraints	Description
doctor_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each doctor
first_name / last_name	VARCHAR(50)	NOT NULL	Doctor full name
specialization	VARCHAR(100)	NOT NULL	Area of medical specialty
department_id	INT	NOT NULL, FOREIGN KEY	Links to departments table
hire_date	DATE	NOT NULL	Date the doctor was hired
salary	DECIMAL(10,2)	NOT NULL	Annual salary
email / phone	VARCHAR	email is UNIQUE	Contact details
is_active	TINYINT	DEFAULT 1	1 means active, 0 means inactive

4.3 patients (Main Table)

This is the main table. It has 120 rows and stores patient personal and registration details.

Column	Type	Constraints	Description
patient_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each patient
first_name / last_name	VARCHAR(50)	NOT NULL	Patient full name
date_of_birth	DATE	NOT NULL	Date of birth
gender	VARCHAR(10)	NOT NULL	Male, Female, or Other
phone / email / address	VARCHAR	Optional	Contact and address details
blood_type	VARCHAR(5)	Optional	A+, A-, B+, B-, AB+, AB-, O+, O-
registration_date	DATE	NOT NULL	Date the patient registered
emergency_contact	VARCHAR(100)	Optional	Name and phone of emergency contact

4.4 appointments

Records visits between patients and doctors. Includes status and notes.

Column	Type	Constraints	Description
appointment_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each appointment
patient_id	INT	NOT NULL, FOREIGN KEY	Links to patients table
doctor_id	INT	NOT NULL, FOREIGN KEY	Links to doctors table
appointment_date	DATE	NOT NULL	Date of the appointment
appointment_time	VARCHAR(10)	NOT NULL	Time in HH:MM format
status	VARCHAR(20)	NOT NULL, DEFAULT Scheduled	Scheduled, Completed, Cancelled, or No-Show
reason / notes	VARCHAR / TEXT	Optional	Reason for visit and clinical notes
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	When the record was created

4.5 bills

Stores billing and payment information for appointments.

Column	Type	Constraints	Description
bill_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each bill
appointment_id	INT	NOT NULL, FOREIGN KEY	Links to appointments table
patient_id	INT	NOT NULL, FOREIGN KEY	Links to patients table
amount	DECIMAL(10,2)	NOT NULL	Amount charged
bill_date / due_date	DATE	bill_date is NOT NULL	Billing date and payment due date
status	VARCHAR(20)	NOT NULL, DEFAULT Pending	Pending, Paid, Overdue, or Cancelled
payment_method	VARCHAR(30)	Optional	Cash, Credit Card, Debit Card, or Insurance
paid_date	DATE	Optional	Date the bill was paid

5. SQL Query Coverage

The project contains more than 60 SQL queries. They cover the topics listed below:

Category	Topics Covered
Basic Queries	SELECT, WHERE, ORDER BY, LIMIT, DISTINCT
Filtering	AND, OR, BETWEEN, LIKE, NOT LIKE, IN, NOT IN, IS NULL, IS NOT NULL
Aggregation	COUNT, SUM, AVG, MIN, MAX with GROUP BY and HAVING
Date Functions	YEAR, MONTH, QUARTER, WEEK
Conditional Logic	CASE statements
Joins	INNER JOIN, LEFT JOIN, RIGHT JOIN, and joins across 3 or 4 tables
Nested Queries	Subqueries in SELECT, FROM, and WHERE clauses

6. How to Use This Project

6.1 Setup Steps

1. Create the database with: `CREATE DATABASE hospital_management_system;`
2. Run the `CREATE TABLE` script to build all five tables.
3. Load the data using the `INSERT` files in this order: departments, doctors, patients, appointments, bills.
4. Run the queries from `Hospital_Queries_MySQL.sql` to practice.

6.2 Notes

- If you need to load data in bulk, you can temporarily turn off foreign key checks with `SET FOREIGN_KEY_CHECKS = 0;` and turn them back on afterward with `SET FOREIGN_KEY_CHECKS = 1;`.
- `NULL` values in optional columns such as phone, notes, and payment_method are normal.
- The `is_active` column on the doctors table uses 1 for active doctors and 0 for inactive ones.

7. Project Files

File	Purpose
01_Create_Database_and_Tables_MySQL.sql	Creates the database and all five tables with keys and constraints
insert_departments.sql	INSERT statements for 50 departments
insert_doctors.sql	INSERT statements for 60 doctors
insert_patients.sql	INSERT statements for 120 patients
insert_appointments.sql	INSERT statements for 150 appointments
insert_bills.sql	INSERT statements for 105 bills
Hospital_Queries_MySQL.sql	More than 60 practice queries covering all required topics